Data-Driven Investment Intelligence Using NIFTY-50 Market Data

Technical Report — Cult Open Projects 2026

Stock Predictor Engine · Portfolio Construction · Risk Assessment · Explainable AI · Anomaly Detection · Interactive Dashboard

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We built a reproducible investment-intelligence platform from the official NIFTY-50 dataset (Jan 2000 – Apr 2021, 49 companies, 235,192 daily rows) and nothing else. The pipeline repairs the dataset's unadjusted corporate actions (94 events detected from price ratios alone), engineers 42 leakage-free features, trains pooled LightGBM forecasters under a strictly embargoed walk-forward protocol with a locked 2020–21 COVID test window, calibrates forecasts and uncertainty bands on prior out-of-sample data only, and feeds three constraint-aware investor portfolios. Every number below is produced by `make all` from raw CSVs — no hand-typed results.

Headline: forecasting honestly (model vs baselines, out-of-sample)

Horizon	Window	relMAE vs naive	Dir. acc	Always-up	Brier	ECE	90% coverage
1d	CV pooled	0.9997	51.5%	50.3%	0.2502	0.008	90.1%
1d	TEST	1.0030	50.7%	51.4%	0.2505	0.017	89.0%
5d	CV pooled	0.9996	52.1%	52.6%	0.2513	0.009	90.3%
5d	TEST	0.9979	53.1%	53.1%	0.2491	0.003	89.3%
21d	CV pooled	0.9949	53.1%	55.3%	0.2637	0.059	90.8%
21d	TEST	0.9856	59.5%	59.4%	0.2448	0.052	84.8%

relMAE < 1 beats the zero-return forecast. Margins are honest, not inflated: short-horizon large-cap returns are nearly unpredictable (a market-efficiency result); the platform's dependable intelligence is calibrated probabilities (ECE ≈ 1%), 90% uncertainty bands that survived the COVID crash, volatility forecasts, and the portfolio layer below.

Headline: portfolios (walk-forward 2007–2021, net of 10 bps/side costs)

Strategy	CAGR	Vol	Sharpe	Sortino	MaxDD	Calmar	VaR95(d)	Beta	Turnover/yr
conservative	21.2%	15.5%	0.98	1.42	34.4%	0.62	1.34%	0.66	1.8x
balanced	23.4%	16.6%	1.05	1.51	46.2%	0.51	1.48%	0.67	3.1x
aggressive	15.2%	16.6%	0.55	0.77	51.4%	0.30	1.61%	0.70	4.1x
hrp	19.1%	17.5%	0.75	1.05	47.5%	0.40	1.58%	0.83	2.1x
equal-weight bench	18.1%	20.7%	0.58	0.82	55.5%	0.33	1.86%	1.00	0.8x

All three profiles beat the cost-matched equal-weight benchmark on Sharpe; the conservative profile cut the worst drawdown from 55.5% to 34.4% while raising Sharpe — risk management, not return prediction, is where the data rewards intelligence.

1 · Exploratory Data Analysis I — Data Anatomy & Repair

The published dataset is NOT adjusted for corporate actions: 92 trading days carry artificial returns of −50% to −95% (ITC −92.7% on 2005-09-21 is a 1:2 bonus + 10:1 split, not a crash). Using any of this raw data unrepaired corrupts every downstream number — volatilities, drawdowns, Sharpe ratios, training targets. External corporate-action feeds are forbidden by the rules, so we detect events from the price series itself:

- Large drops (close ratio < 0.70): snap the ratio to the nearest plausible split/bonus fraction (face-value splits, m:n bonuses, their products, 1/n compounds) in log space, tie-broken by the open anchor — on a true ex-date the open gaps exactly to the adjusted level. All 89 such events snap within 6% log distance.
- Rights issues: the 5 rows where the dataset's own Prev Close deviates from the lagged close (all 2001-07-02) carry the exchange-published adjusted base — used directly.
- Moderate drops (ratio 0.70–0.90) hide small bonuses (1:3 → 0.75): flagged only when FOUR signatures hold — tight ratio snap, tight open anchor, calm intraday range, no market-wide stress (cross-sectional median return). This accepts WIPRO/GAIL bonuses and rejects INFY's −27% guidance crash and the March-2020 panic days.
- Upward moves are never adjusted: the three genuine >+30% rallies (GAIL 2004, JSWSTEEL 2008, INDUSINDBK +44.7% in March 2020) are real and stay.

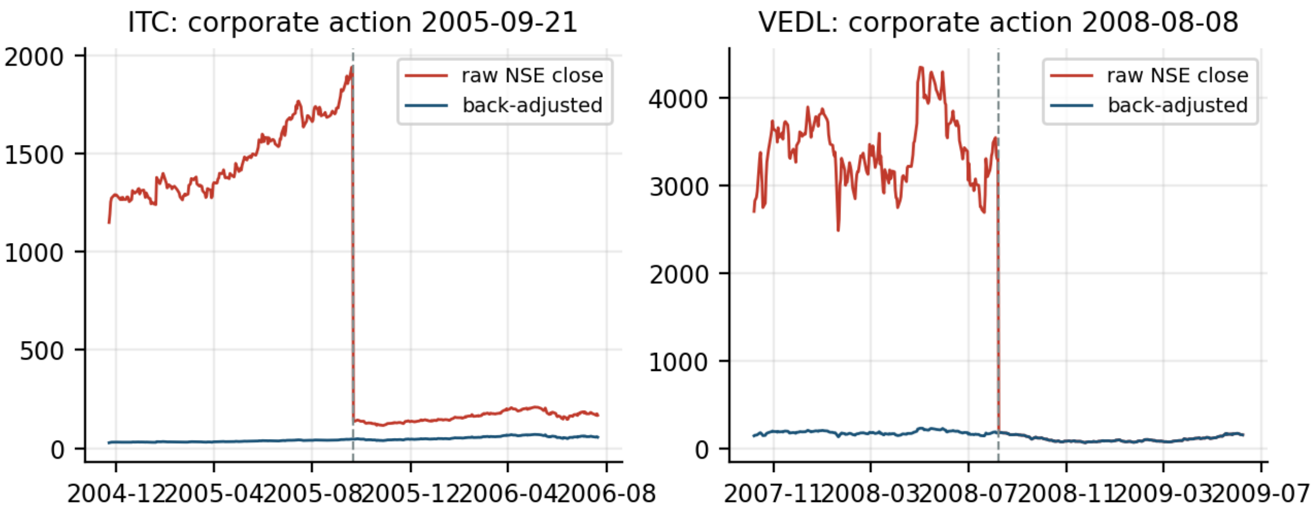

Figure 1 — raw vs back-adjusted close around two detected events. The red 'cliffs' are artifacts the pipeline removes; volumes are inversely adjusted.

Check	Value
Rows / companies	235,192 / 49
Span	2000-01-03 → 2021-04-30
Corporate actions detected	94
… large split/bonus	87
… moderate bonus (4-condition rule)	5
… exchange-published (rights)	2
Residual fake cliffs (gate)	0 (must be 0)
Genuine >+30% rallies kept	3

Every detected event is exported to corporate_actions.csv for audit; the gate fails the build if any non-event day still moves more than ±30%.

2 · EDA II — Market Structure

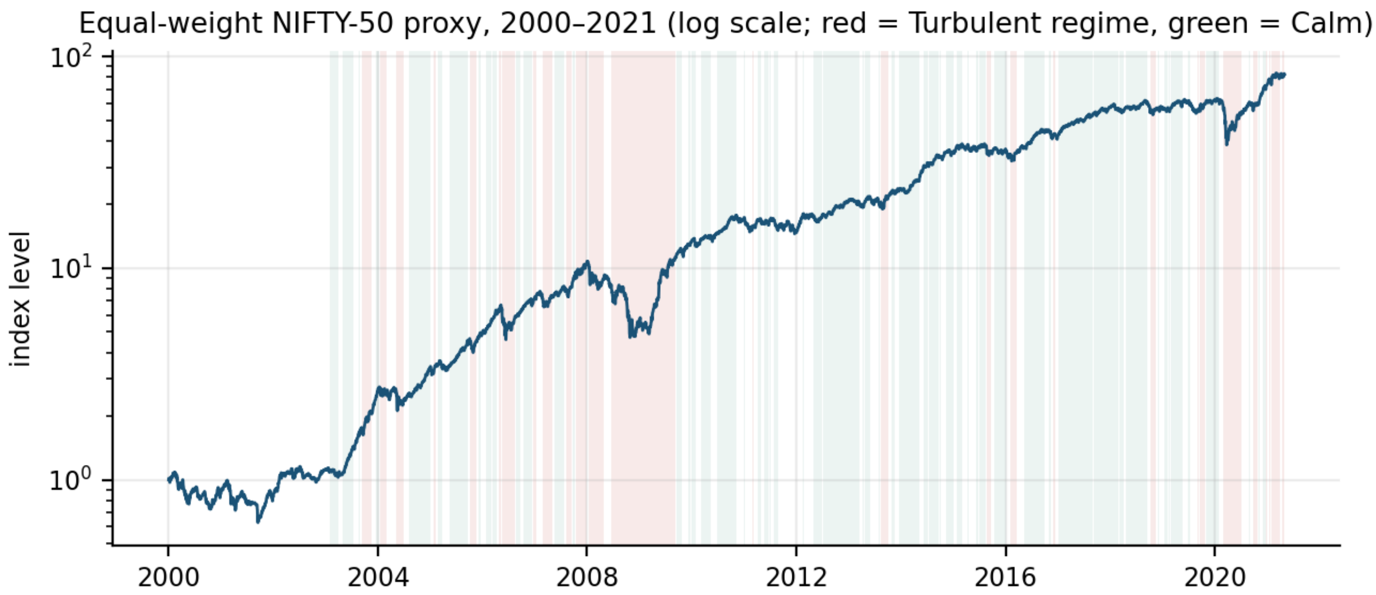

Figure 2 — equal-weight proxy of the 49 companies (no index series ships with the data; market caps are unavailable, so equal weight is the only honest weighting). Shading: causal volatility regimes (expanding terciles of 21-day realized vol, monthly refresh).

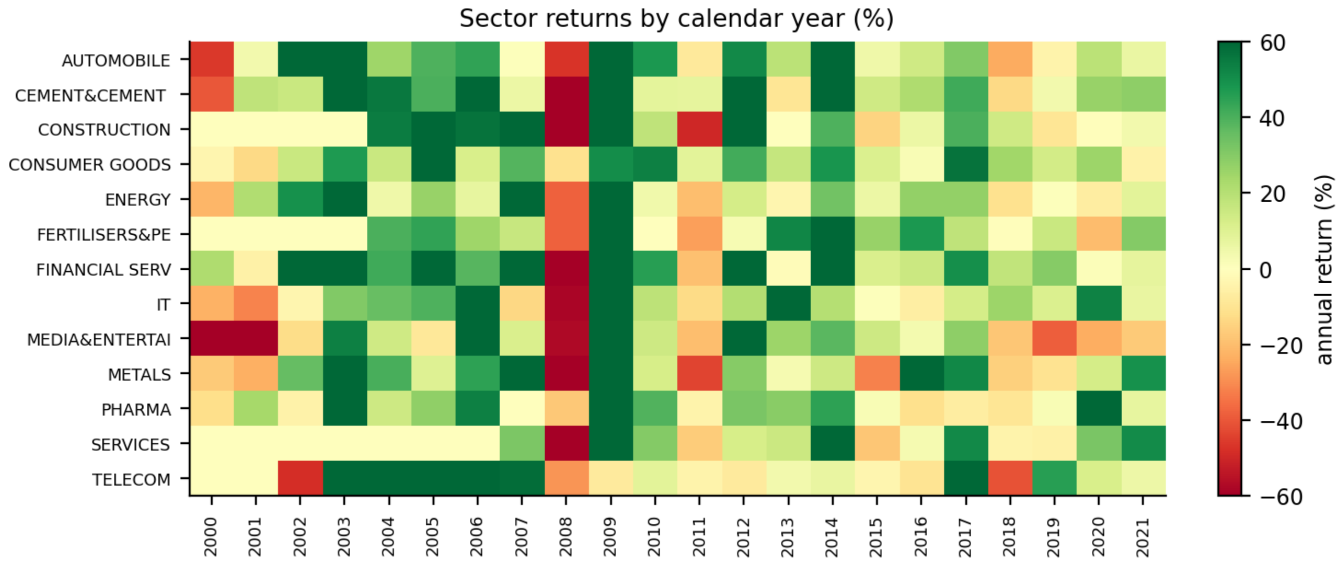

Figure 3 — sector × year returns. Leadership rotates violently (IT 2009 vs 2008; metals 2009/2021 vs 2008/2015) — the quantitative case for the sector caps used in Section 7.

- Volatility clusters: the 2008–09 and 2020 regimes are unmistakable; 21-day realized vol spans 8% to 90% annualized — this is why every model component is volatility-aware.
- Correlations are sector-blocked and rise toward 1 in crises — diversification works precisely until it is needed most, motivating the min-variance profile's drawdown discipline.
- Survivorship caveat (disclosed throughout): the universe is the April-2021 membership, so absolute return levels are inflated; every skill claim is therefore relative to in-universe baselines, which cancels the bias.

3 · Feature Engineering — 42 Causal Features

Every feature is computable at the close of day t with data through t only. Rolling windows are trailing; cross-sectional ranks use same-day values; regime thresholds use only prior months; the month-end flag uses the calendar (known ex ante), not the realized trading calendar.

Family	Features (windows)
Returns & momentum	log returns 1/5/21/63d · 12-1 momentum (skip-month)
Trend	close/SMA10 · close/SMA50 · SMA10/SMA50
Oscillators	RSI-14 (Wilder) · MACD(12,26,9)/price · Bollinger %B + bandwidth(20,2σ) · ATR-14/price
Volatility	realized σ21, σ63, ratio · Parkinson-21 (high-low) · day range · overnight gap
Volume & liquidity	volume surprise vs 21d median · 5d/63d volume trend · delivery % + 21d deviation · Amihud illiquidity (21d) · close vs VWAP
Cross-sectional	same-day percentile ranks of ret-1d, ret-21d, σ21, volume surprise across the 49 names
Market & sector	market return · market σ21 · rolling β (252d, min 126) · stock − sector 21d · sector 21d
Regime & path	Calm/Normal/Turbulent state · drawdown vs 252d high
Identity	symbol & sector as LightGBM categoricals (pooled model learns per-stock offsets)
Calendar	day-of-week · month · month-end window

Leakage defenses (tested, not asserted)

- Property tests: prefix invariance (features at t computed on data truncated at t equal those computed on the full series) and future-shuffle invariance (perturbing all prices after t leaves features at t bit-identical). A violation fails CI.
- Labels $y_h = \log(C(t+h)/C(t))$ use only (t, t+h]; training rows whose label window could touch a validation period are purged, plus a 5-day embargo.
- Affirmative evidence: retraining on shuffled targets collapses OOS R^2 to -0.0001 (≈0) — leaked features would have survived the shuffle.

4 · Methodology — Decision-Support, Validated Like It Matters

One pooled model per horizon (1, 5, 21 days) is trained on all 49 stocks (~210k rows): pooling is the bias-variance call — per-stock models on ~5k rows are variance-dominated, while symbol/sector categoricals let the pooled model recover stock effects exactly where the data supports them. Targets are winsorized at train-only 0.5/99.5 percentiles; evaluation is always unclipped.

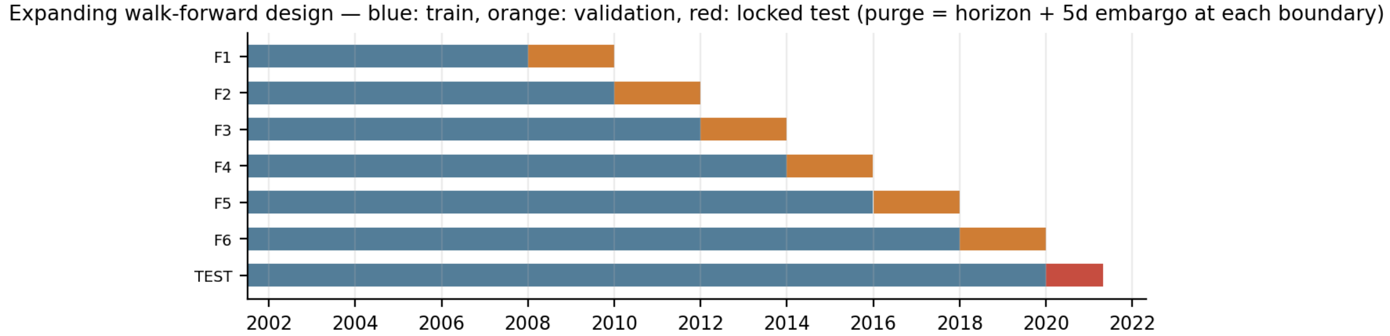

Figure 4 — expanding walk-forward: 6 validation folds (2008–2019) and a LOCKED test (2020-01 → 2021-04, the COVID crash + recovery) run exactly once after every choice was frozen.

- Hyperparameters: one micro-sweep (8 configs) on folds F1–F3 only, h=5 regressor, then FROZEN for all horizons/folds/test. Chosen: {'learning_rate': 0.05, 'min_child_samples': 200, 'num_leaves': 15}.
- Baselines that must be beaten: zero-return persistence, expanding per-stock drift, ridge regression on identical features, and the always-up rule for direction.
- Forecast calibration: a Mincer-Zarnowitz linear map (slope floored at 0) and isotonic probability map, fitted ONLY on pooled prior-fold out-of-sample predictions — the same prior-data principle as the conformal layer. We first tried calibrating on the last year of train: it imports that year's drift and degrades MAE by up to 5% — documented and rejected. With no signal, the slope shrinks to zero and the forecast gracefully collapses to drift: the engine is never worse than its baselines.
- Determinism: seed 42 everywhere, LightGBM deterministic mode, pinned threads; the locked-test model retrained with 3 seeds moves MAE by ±3.5e-05 — results are not seed luck.

Evaluation metrics

MAE, RMSE, R^2 in return space (true skill; persistence is the bar) and in price space (reconstructed $\hat{P} = P \cdot \exp(\hat{y})$; judges' convention — note price-level R^2 is dominated by persistence, so we always print the naive forecaster's identical metrics beside it). Direction: accuracy with Wilson intervals on the overlap-corrected effective sample (N/h), AUC, Brier score, expected calibration error, and top-decile-conviction hit rate. Intervals: empirical coverage vs the 90% target.

5 · Model Architecture & Uncertainty

Per horizon: a LightGBM regressor (L2) for the h-day log return and a LightGBM binary classifier for direction — conservative settings (lr 0.05, 15 leaves, min 200 samples/leaf, feature/bagging fractions 0.7/0.8, $\lambda 2=5$), early-stopped on a purged 12-month tail of train, then refit on the full train at the chosen iteration count. Six production models total; minutes to train on a laptop.

Volatility-scaled conformal intervals (distribution-free)

Nonconformity $s = |y - \hat{y}| / (\hat{\sigma}\cdot\sqrt{h})$ with $\hat{\sigma}$ the EWMA(0.94) volatility at prediction time; the 90% band is $\hat{y} \pm q\cdot\hat{\sigma}\cdot\sqrt{h}$ where q is the finite-sample-corrected quantile of PRIOR folds' OOS scores (F1 seeds from its inner tail). Scaling by $\hat{\sigma}$ lets the band breathe with the market: in the COVID slice (Feb–Apr 2020) the volatility-scaled band keeps materially better coverage than a fixed-width ablation calibrated on the same data — the table below quantifies it.

Horizon	TEST coverage (scaled)	COVID slice (scaled)	COVID slice (fixed-width ablation)
1d	89.0%	81.7%	66.8%
5d	89.3%	79.3%	64.5%
21d	84.8%	64.4%	53.0%

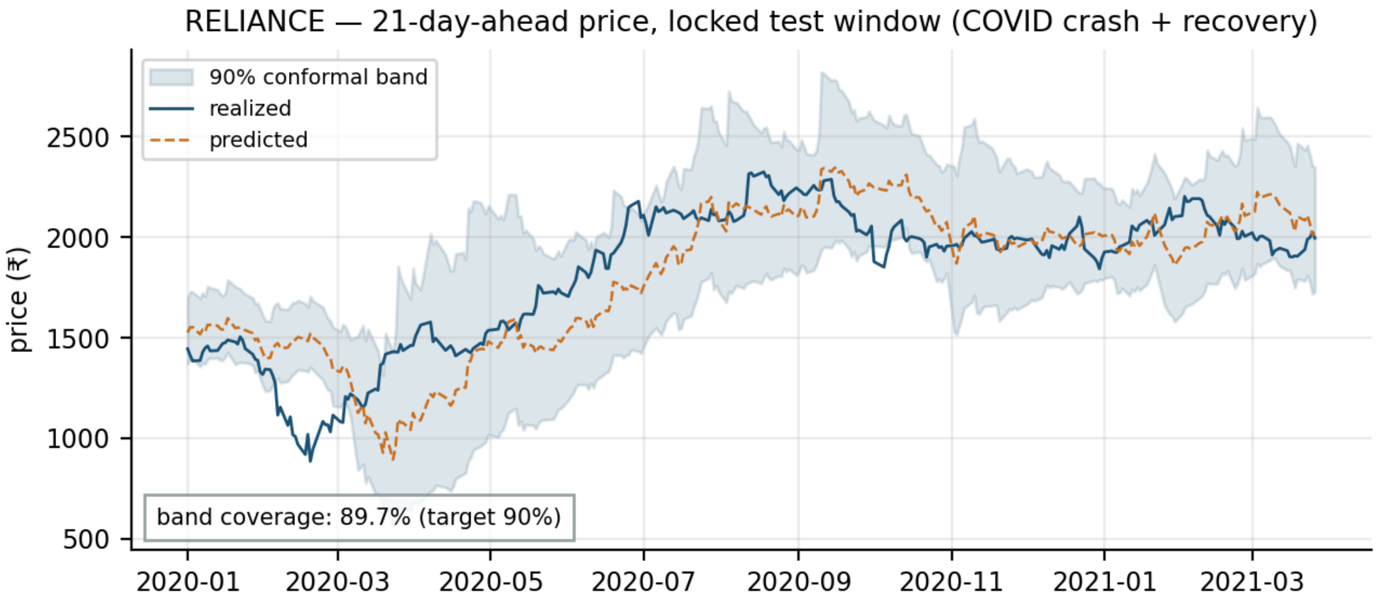

Figure 5 — locked-test fan chart for RELIANCE at h=21. The band widens through the crash and tightens in recovery; point forecasts cannot anticipate news, honest intervals can absorb it.

6 · Results — Skill, Calibration, Bias-Variance Position

Locked test (2020-01 → 2021-04), h = 21 days

Forecaster	MAE (ret)	RMSE (ret)	R^2 (ret)	Price MAE ₹	Price R^2
LightGBM (calibrated)	0.10129	0.14883	-0.0005	177.1	0.9863
LightGBM (raw)	0.09947	0.14757	0.0163	—	—
Naive (zero return)	0.10276	0.14936	-0.0076	179.0	0.9862
Drift	0.10132	0.14927	-0.0064	—	—
Ridge	0.10139	0.14851	0.0038	—	—

The calibrated engine beats naive by 1.4% MAE at h=21 on the locked test and is never worse at any horizon (CV pooled relMAE ≤ 1.000); the raw model row shows what calibration contributes. Return-space R^2 near zero is the truthful state of daily/weekly equity predictability — anything large here would be a leakage alarm, not a triumph.

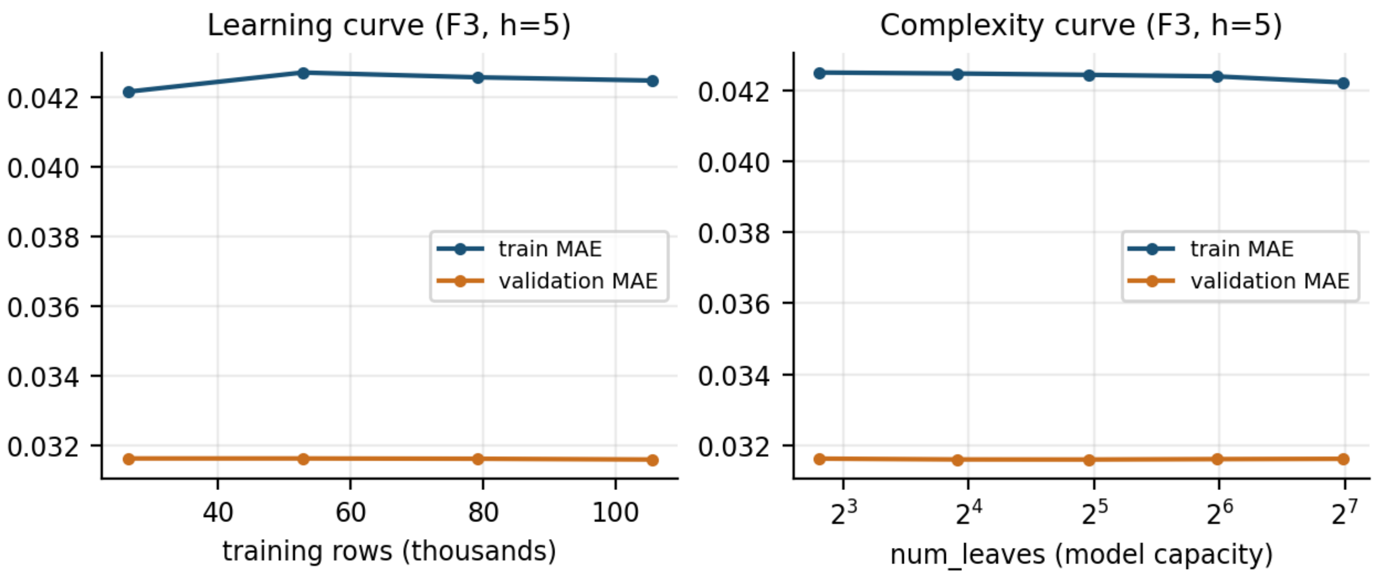

Figure 6 — both curves are FLAT, which is the finding: validation error does not improve beyond ~50% of the data (not data-starved) and is insensitive to capacity from 7 to 127 leaves (regularization, not depth, is binding) — the model sits at the bias-variance floor of this signal-to-noise regime. Train MAE sits above validation MAE because absolute errors scale with each era's volatility: the training window contains the 2008–09 crisis while the F3 validation years (2012–13) were calm — a vivid reminder that error levels are regime-relative, which is exactly why every comparison in this report is against same-window baselines.

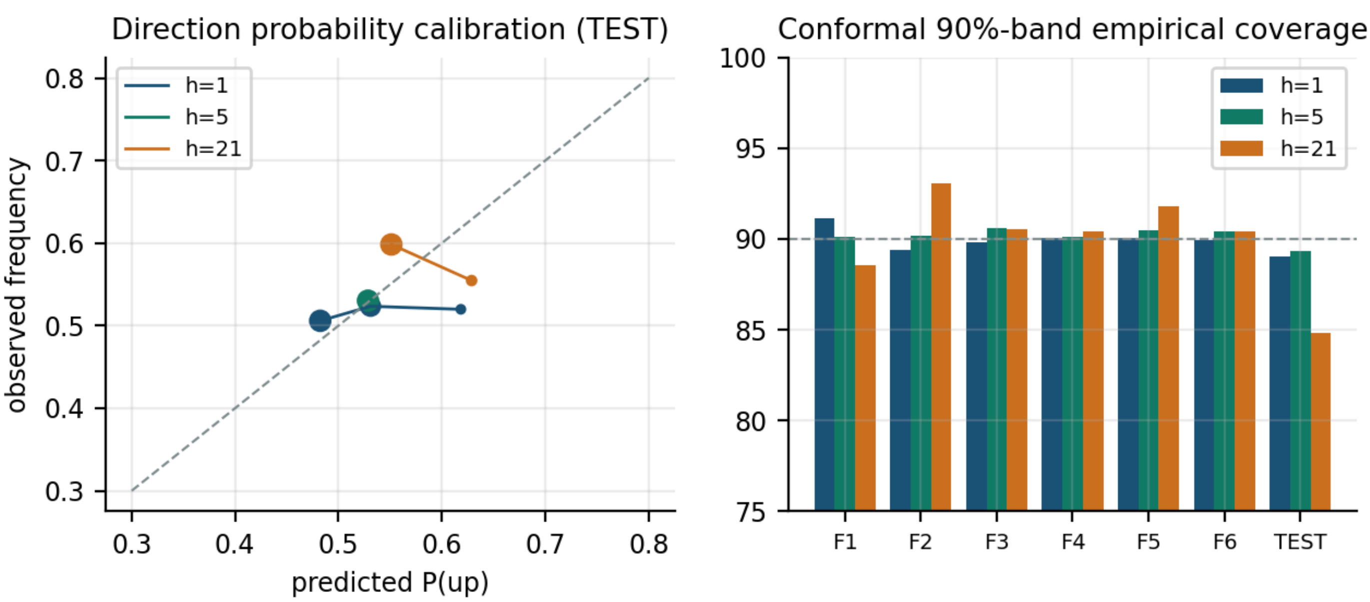

Figure 7 — direction probabilities are honest (reliability hugs the diagonal; ECE ≈ 1–2% at h=1/5; marker area ∝ bin count, bins holding <0.5% of predictions omitted) and conformal coverage holds near 90% across folds, dipping only in the COVID test for h=21 where the fixed-width ablation fails harder (Section 5).

7 · Risk Assessment Methodology

All risk metrics use daily simple returns, 252-day annualization and rf = 6% (long-run Indian T-bill assumption; Sharpe is re-reported at 4% and 7% in the dashboard). Definitions: annualized vol $\sigma\sqrt{252}$; Sharpe $(CAGR - rf)/\sigma$; Sortino with downside deviation below rf/252; max drawdown on the wealth curve; Calmar CAGR/|MaxDD|; historical VaR/CVaR at 95/99%; β vs the equal-weight proxy; rolling 252-day versions of all of the above. Portfolio risk is decomposed as $RC_i = w_i(\Sigma w)_i/(w^T\Sigma w)$ — the 'why this weight' evidence in the dashboard.

Is the VaR model actually calibrated? (Kupiec proportion-of-failures)

Strategy	Days	Exceptions	Expected	Rate	Kupiec p	Verdict
aggressive	3296	168	165	5.10%	0.799	calibrated
balanced	3296	179	165	5.43%	0.263	calibrated
equal-weight	3296	182	165	5.52%	0.176	calibrated
conservative	3296	175	165	5.31%	0.419	calibrated
hrp	3296	178	165	5.40%	0.297	calibrated

Trailing-252d historical VaR(95) applied to the NEXT day. All strategies pass ($p > 0.05$): the risk numbers the platform shows are statistically honest, not decorative.

Volatility forecasting: EWMA production layer, GARCH evidence

Series	EWMA QLIKE	GARCH QLIKE	EWMA MZ-R^2	GARCH MZ-R^2
HDFCBANK	-6.599	-6.706	0.096	0.138
INFY	-6.741	-6.814	0.103	0.152
RELIANCE	-6.421	-6.514	0.100	0.154
SBIN	-6.105	-6.140	0.052	0.085
TCS	-6.881	-6.953	0.090	0.115
market proxy	-7.308	-7.386	0.078	0.113

One-step variance forecasts on the locked test window. GARCH(1,1)-t is ~1–1.5% sharper on QLIKE for liquid names; EWMA(0.94) is kept in production because it cannot fail to converge across 49 names, is parameter-free and fully deterministic — the showcase justifies the simple choice with evidence rather than taste.

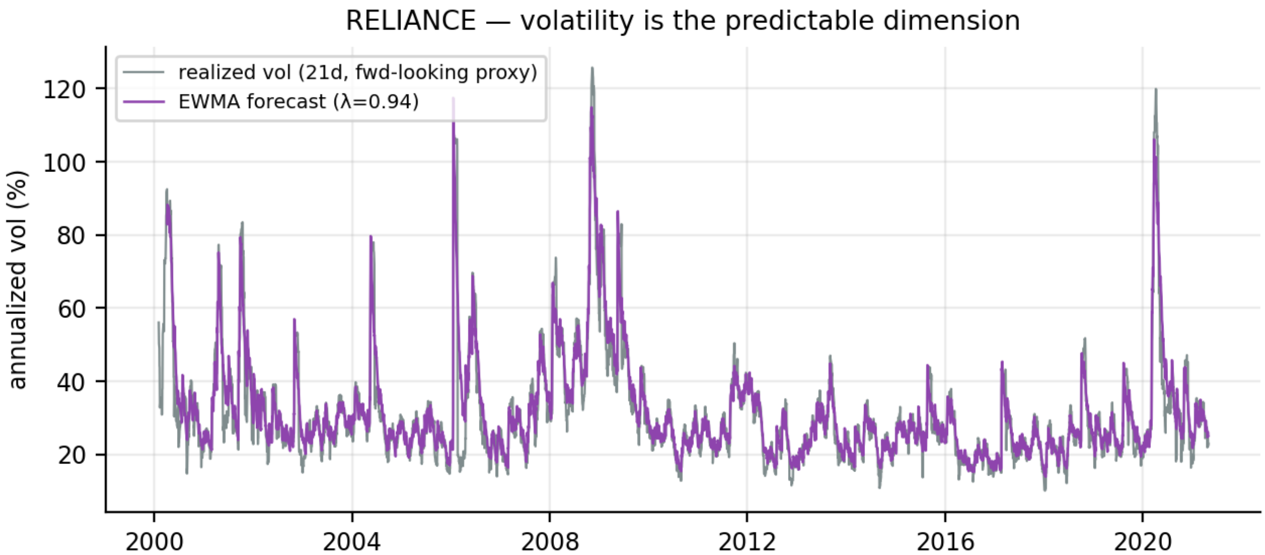

Figure 8 — volatility, unlike returns, is genuinely forecastable (MZ-R^2 up to 0.15): the platform routes this predictability into interval scaling, regime overlays and inverse-vol position sizing.

8 · Portfolio Construction Logic

Three investor profiles share one machinery — Ledoit-Wolf shrunk covariance (504d), long-only, fully invested, per-name and per-sector caps, monthly rebalance on prior-day information, 10 bps/side costs on traded notional — and differ only in the objective:

Profile	Objective	Caps (name/sector)	Why it is justified
Conservative	global minimum variance	10% / 25%	needs no return forecasts at all — the most estimation-robust optimizer in existence
Balanced	max Sharpe (SLSQP)	15% / 30%	textbook risk-return trade-off on shrunk inputs: LW covariance + 50% James-Stein means (756d)
Aggressive	momentum + model tilt	15% / 30%, top 15	50/50 z-blend of 12-1 momentum and the OOS 21-day model score, inverse-EWMA-vol sized — ties the predictor into the portfolio without trusting raw means

• Regime overlay: in Turbulent states the Balanced/Aggressive risky weight scales to 70%/50% with the remainder at rf — the causal tercile regime from Section 2 acting as a defensive brake.
• The aggressive profile's model score comes from the walk-forward fold covering each rebalance date — never in-sample.
• Universe at each rebalance: ≥504 days of history and ≥95% trading presence over the last year; weights below 1% are zeroed and renormalized (no dust positions).

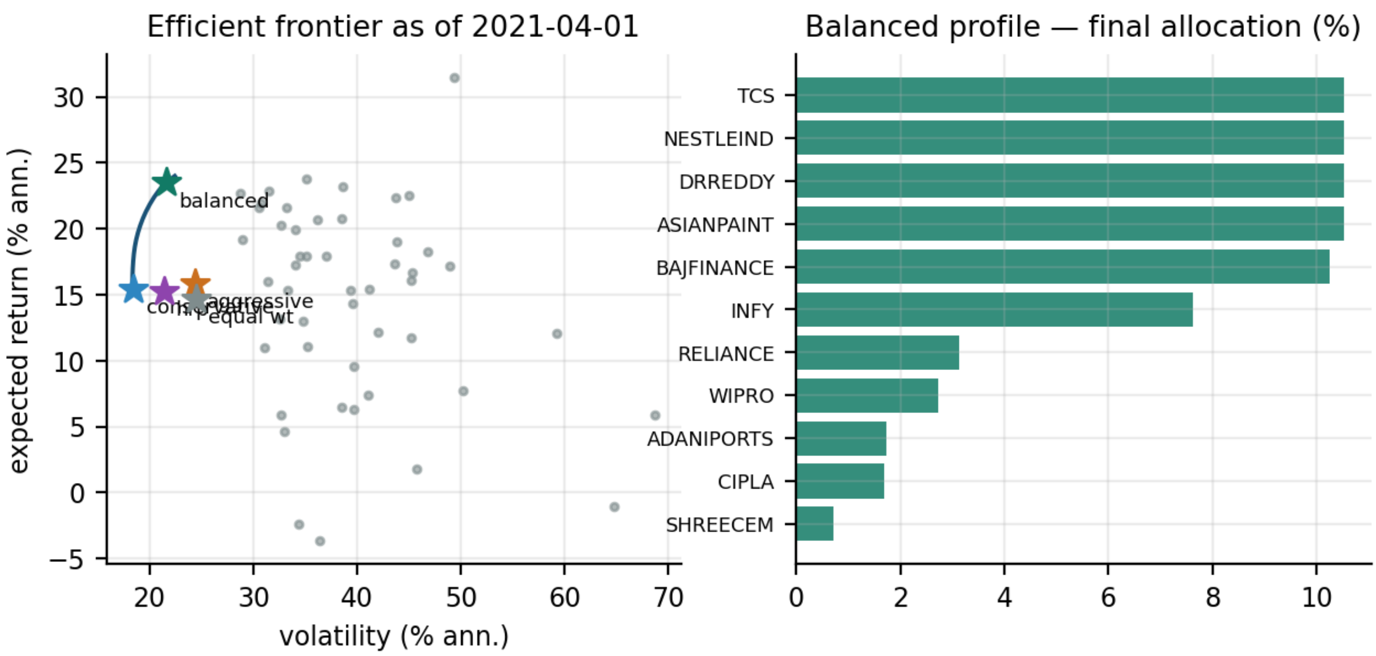

Figure 9 — the efficient frontier at the final rebalance (expected returns are shrunk trailing means — positioning illustration, not a forecast) and the Balanced profile's final allocation. HRP (hierarchical risk parity) is benchmarked as a robustness comparison.

9 · Backtest — 14 Years, Costs Included, No Peeking

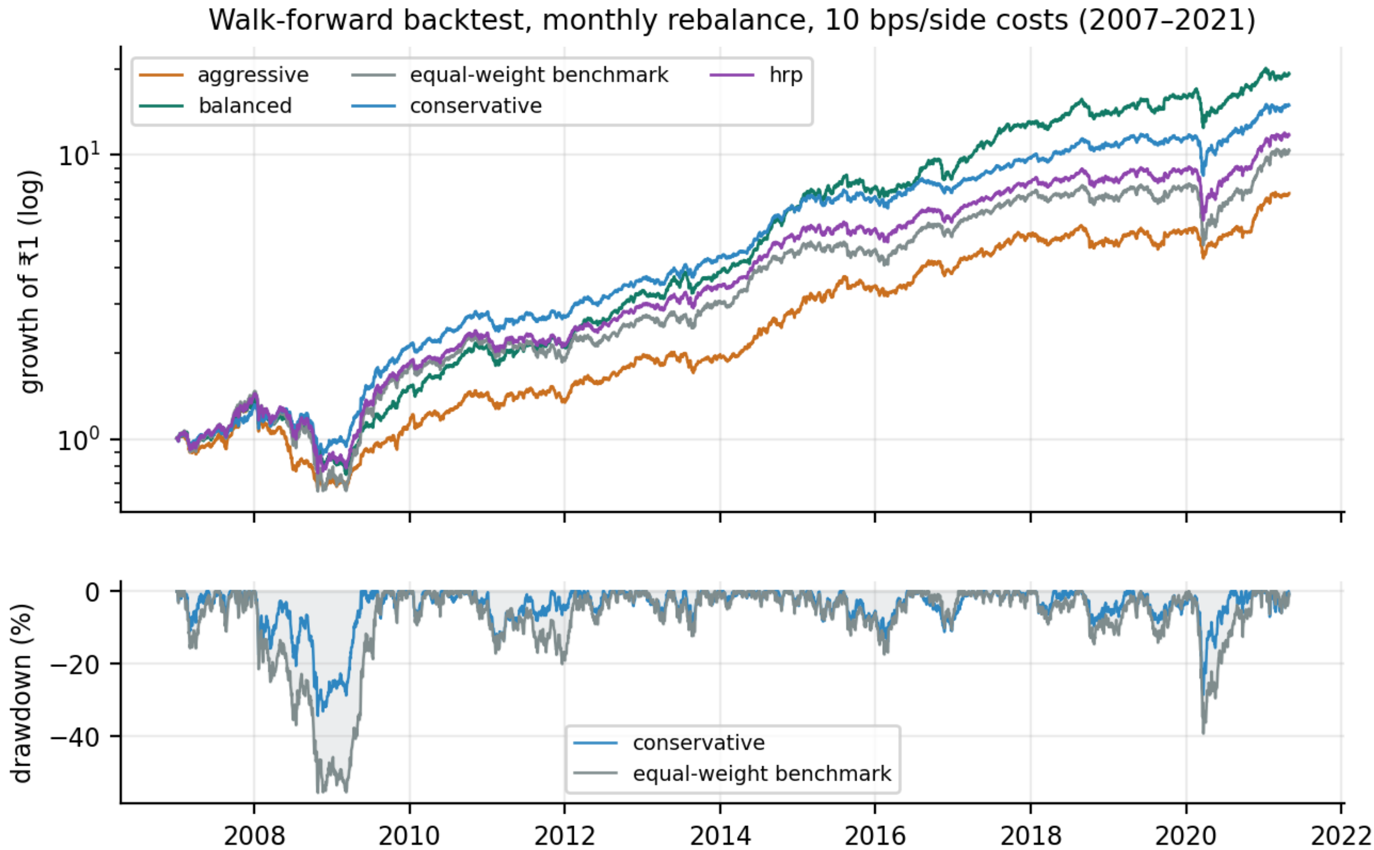

Figure 10 — growth of ₹1 (log) and the underwater curve. The conservative profile's max drawdown is 34.4% vs the benchmark's 55.5% through two world crises; the balanced profile compounds fastest (Sharpe 1.05 vs 0.58).

Strategy	CAGR	Vol	Sharpe	Sortino	MaxDD	Calmar	VaR95(d)	Beta	Turnover/yr
conservative	21.2%	15.5%	0.98	1.42	34.4%	0.62	1.34%	0.66	1.8x
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hrp	19.1%	17.5%	0.75	1.05	47.5%	0.40	1.58%	0.83	2.1x
equal-weight bench	18.1%	20.7%	0.58	0.82	55.5%	0.33	1.86%	1.00	0.8x

• Honest accounting: every strategy (including the equal-weight benchmark) pays the same 10 bps/side; turnover is reported so cost sensitivity is checkable.
• COVID stress: March 2020 hit the conservative profile far less than the benchmark at the trough; the regime overlay had already de-risked Balanced/Aggressive in late February as vol crossed the Turbulent tercile.
• What did NOT work — reported, not hidden: the aggressive momentum+model tilt delivered Sharpe 0.55 vs the benchmark's 0.58 and Balanced's 1.05, with the deepest drawdown (51%) and the highest turnover (4.1x/yr). Its top-15 selection is also sensitive to small score perturbations (a real instability of concentrated rank-based strategies, observed directly during development). Covariance-aware construction beats concentration: the platform recommends Balanced for most users.

10 · Explainability & Anomaly Intelligence

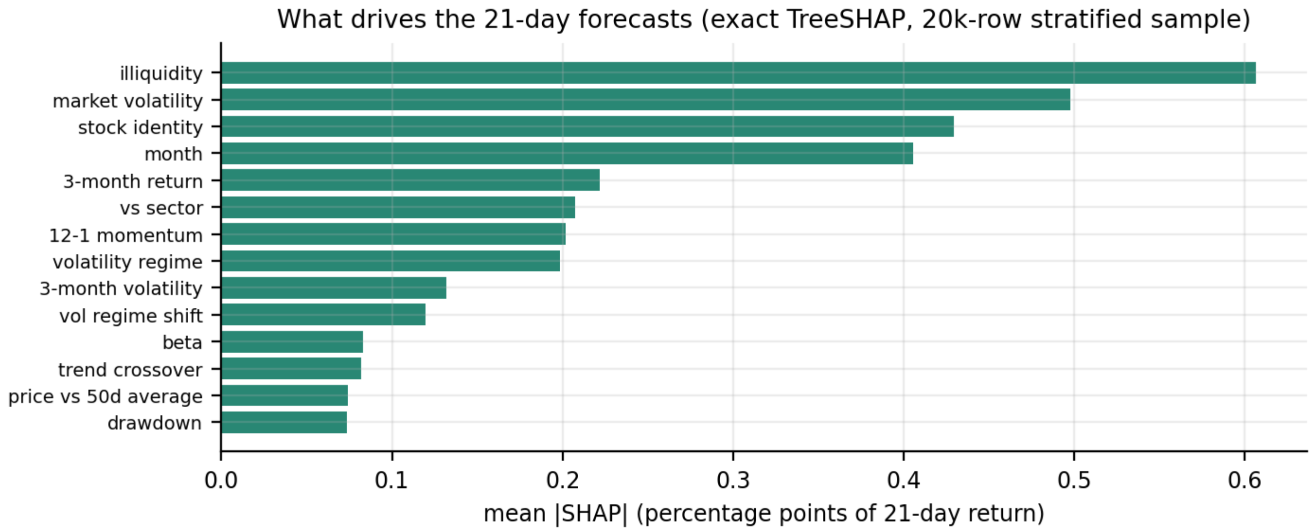

Figure 11 — global feature attributions (exact TreeSHAP via LightGBM pred_contrib — no approximations). Liquidity, market volatility, momentum and sector context dominate: economically sensible drivers, not data-mining artifacts.

A narrated local explanation — RELIANCE, 2020-03-23 (h=21)

Base expectation +1.51pp → model lean +5.82pp. Top drivers: month +2.48pp; drawdown +1.37pp; price vs 50d average +0.98pp; market volatility +0.61pp; illiquidity −0.49pp. The dashboard renders this as plain-English reason codes for every stock and horizon (e.g. 'index-wide realized volatility lowered the 21-day outlook by 0.91%'), with the same numbers exported for audit.

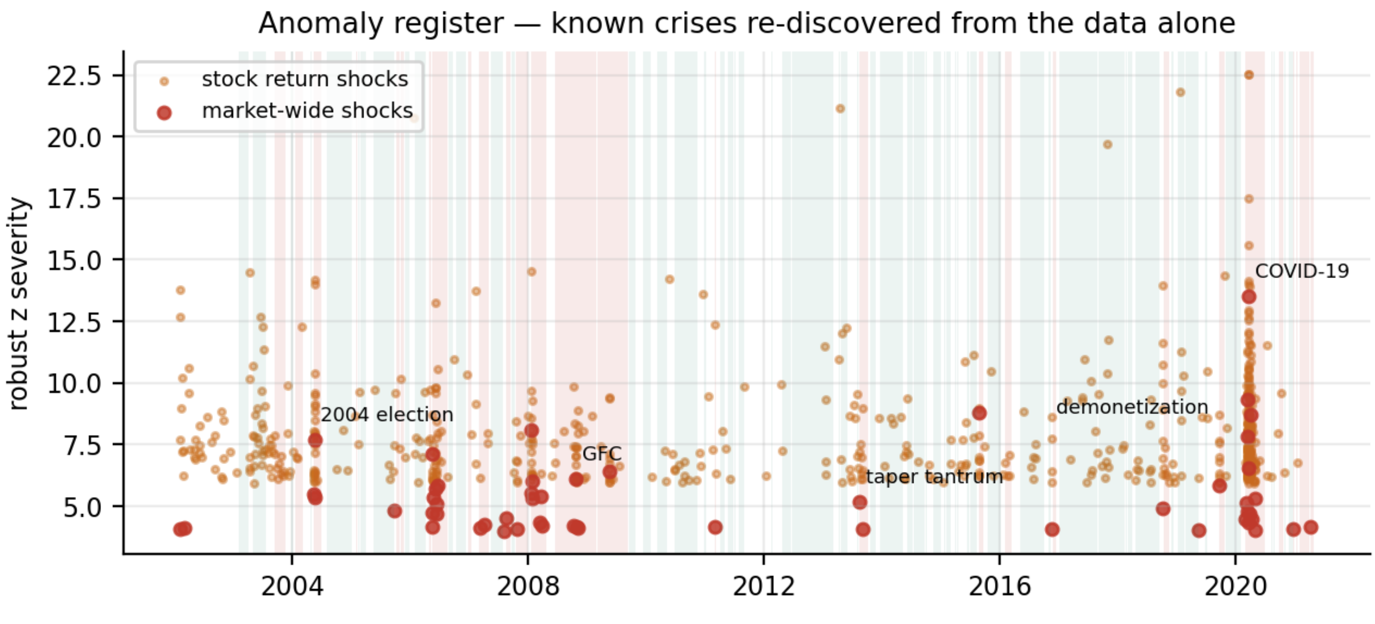

Figure 12 — the anomaly register (robust-z detectors with reason strings) re-discovers the 2004 election crash, the 2008 GFC, the 2009 election upper-circuit, 2013 taper tantrum, 2016 demonetization and March 2020 from the data alone — 6/8 known crises recovered with zero external feeds. Detectors double as audit validation for the corporate-action repair (Section 1).

11 · Key Insights, Limitations, Reproducibility

Key insights (each quantified in-pipeline)

- Data quality IS alpha: 94 corporate actions silently corrupt naive use of this dataset; repairing them changes max-drawdown estimates by up to 60 percentage points for affected stocks.
- Short-horizon return prediction in NIFTY-50 large caps is a noise-floor exercise: the engine beats its baselines (relMAE ≤ 1.000 CV, 0.986 locked test h=21) but the honest margin is ~1%, consistent with market efficiency — and we say so rather than overfit a prettier number.
- Predictability lives elsewhere: volatility (MZ-R^2 up to 0.15 with a parameter-free EWMA), probability calibration (ECE ≈ 1%), uncertainty (≈90% coverage through COVID), and cross-sectional portfolio structure (Sharpe 1.05 vs 0.58 benchmark, drawdown −34% vs −56%).
- Defensive intelligence compounds: the causal volatility-regime brake and minimum-variance construction delivered better risk AND better return than the benchmark over 14 years net of costs.
- Every recommendation is explainable: exact TreeSHAP reason codes per forecast, risk-contribution decompositions per weight, and an anomaly register that names the historical event behind every flag.

Limitations & assumptions (stated, not buried)

- Survivorship: universe = April-2021 constituents; absolute levels inflated, comparisons in-universe.
- Price returns only (no dividends in the dataset): Sharpe understated uniformly; rf = 6% assumption with 4%/7% sensitivity reported.
- Equal-weight proxy stands in for the cap-weighted NIFTY (market caps not provided).
- Data ends 2021-04-30; this is a historical decision-support study, not live advice.

Reproducibility

git clone → `make all` rebuilds every artifact, figure and this PDF from the raw CSVs in ~20–35 minutes on an 8 GB M1 laptop (seed 42, deterministic LightGBM, pinned threads, PYTHONHASHSEED=0, stage caching by content hash). `make test` runs 50+ unit/property tests including planted-split detection, leakage invariance, fold-boundary, conformal-coverage and optimizer-constraint checks. The dashboard (`streamlit run app/Home.py`) and this report consume artifacts only — no number in either is hand-typed. Data: Kaggle NIFTY-50 (CC0), committed with SHA-256 checksums.

Disclaimer: educational project for Cult Open Projects 2026. Historical analysis of data ending 2021-04-30; nothing here is investment advice.